

Flows, Strategies and an Experiment, End to End

Two flows in production today, a third built by deploying new strategies at two decision points, an experiment between them, and six transactions traced through — with the records as they actually appear. Written to be reviewed line by line with product and engineering.

Version	1.0
Worked example	Token Performance
Findings	4 blocking, 3 advisory
Companion to	Experimentation PRD v1.0

Contents

A0	What this example is for
A1	Decision point registry
A2	Current state — F1 and F2
A3	What D4 actually gates
B1	The two new strategies
B2	F3 as sketched — finding 1
B3	Four ways to fix it
B4	F3 and F4 compiled
C1	The assignment point
C2	Transaction selection — finding 2
C3	Unit of randomisation — finding 3
C4	What 95/5 costs — finding 4
C5	The experiment record
C6	What registration rejects
D1	Six transactions
D2	The records
D3	How the data appears
D4	Readout at day 37
E1	Review agenda

What this example is for

This traces one concrete change through the whole mechanism: two flows that exist today, a new strategy deployed at a decision point, a flow built around it, an experiment between old and new, and the data that comes out the other side. It exists so the team can argue with something specific rather than with an abstraction.

It also surfaces four things that would have broken the experiment if it had been built as first drawn. Each is called out where it arises, and each has a fix. Three are cheap. One changes how long the experiment has to run.

#	Finding	Severity	Where
1	The new flow changes two decision points at once, so a win cannot be attributed to either	Blocking	§B2
2	Selection criteria do not require the treatment credential to exist, so part of the test arm cannot be treated	Blocking	§C2
3	No unit of randomisation declared; the obvious choice is wrong for this decision point	Blocking	§C3
4	95/5 needs 3.0M eligible transactions to detect 25 bps — 5.3× what an even split needs	Blocking on duration	§C4
5	No primary metric, guardrails, sample size, stopping rule or expiry	Advisory	§C6
6	Assignment placed after D4 — correct, and it corrects the PRD	Advisory	§C1
7	D4.1 and D4.2 are labelled inconsistently between the current-state and new-flow notes	Advisory	§A1

Decision point registry

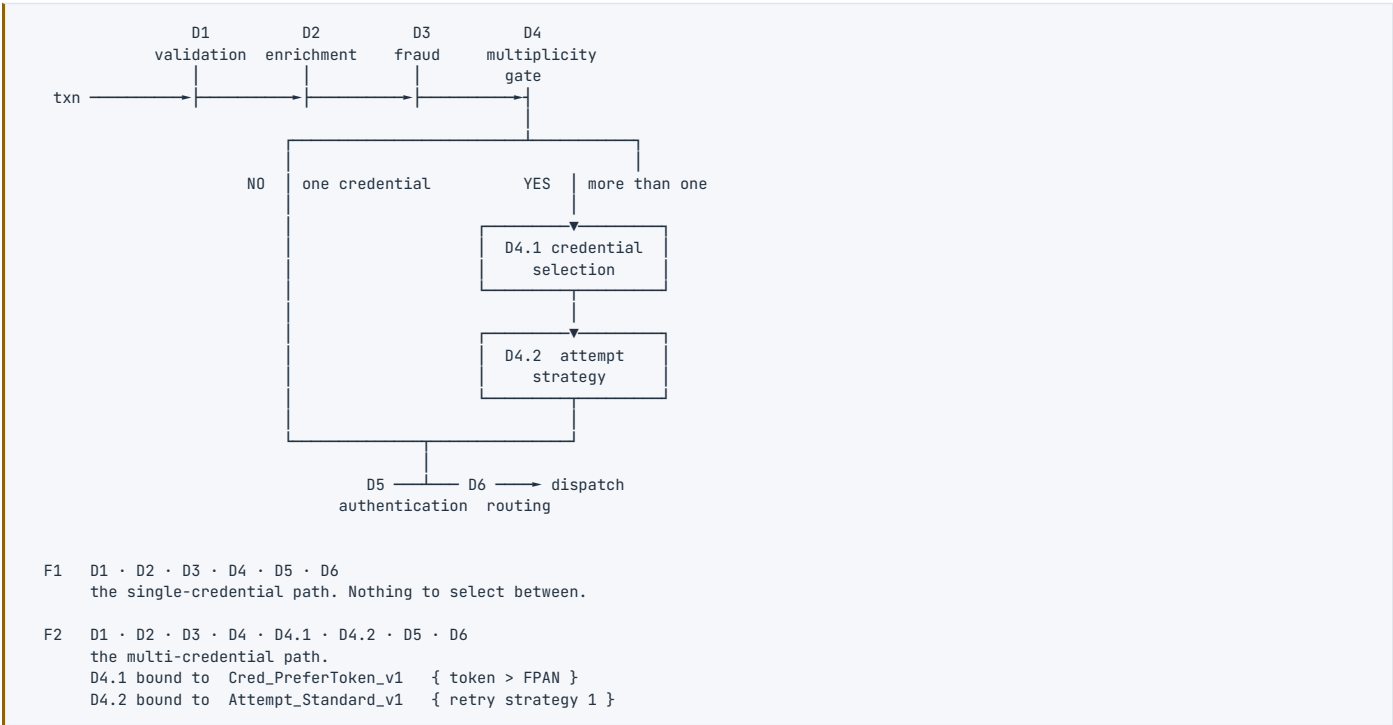
Eight decision points across the two live flows. D4.1 and D4.2 are children of D4 — they execute only on its positive branch.

ID	Decision point	Decides	Experimentable	Layer salt
D1	Ingress validation	Schema, idempotency, authentication of the caller. Not a choice	No	—
D2	Instrument enrichment	Vault response, BIN attributes, credential set. Not a choice	No	—
D3	Fraud evaluation	Risk prediction vector	No — risk owns its own promotion path	—
D4	Credential multiplicity gate	Whether credential selection is needed at all. Deterministic on the vault response	No	—
D4.1	Credential selection	Which credential to present	Yes	salt_cred
D4.2	Attempt strategy	What to try next after a non-approval	Yes	salt_attempt
D5	Authentication	None, exemption, 3DS frictionless, challenge	Yes	salt_auth
D6	Routing	Processor, acquirer, network, message model	Yes	salt_route

Finding 7 — a labelling inconsistency worth settling first

The current-state notes read **D4.1 = credential selection** and **D4.2 = retry**. The new-flow notes attach the FPAN strategy to D4.2 and the retry strategy to D4.1 — the two are swapped relative to the first list. This document uses the first list throughout: **D4.1 is credential selection, D4.2 is attempt strategy**. Confirm that before anything is registered, because the identifiers end up in strategy names, experiment layers, decision records and every readout, and they are expensive to rename once traffic has run under them.

Current state — F1 and F2



Flow	D1	D2	D3	D4	D4.1	D4.2	D5	D6
F1_Single_v4	std	std	std	std	—	—	Auth_RiskAdaptive_v3	Route_Balanced_v7
F2_Multi_v6	std	std	std	std	Cred_PreferToken_v1	Attempt_Standard_v1	Auth_RiskAdaptive_v3	Route_Balanced_v7

What D4 actually gates

D4 asks one question of the vault response: did it return more than one usable credential for this customer and merchant. That question has three possible homes, and the choice matters because it determines where the experiment can be evaluated.

Reading	Consequence	Assessment
A — runtime gate inside one flow D4.1 and D4.2 are conditional nodes that execute only on the positive branch	One flow definition covers both populations. Assignment must be deferred to the gate's outcome	Adopted. Matches the sketch and keeps one flow per txn type
B — two flows selected at runtime F1 and F2 are separate definitions and the flow is re-resolved at D4	Flow resolution happens twice per transaction, and the ExecID recorded at the start is not the one that ran	Rejected. A transaction must carry one ExecID from resolution to record
C — multiplicity becomes part of the txn type Resolved at context assembly, before D1	F1 and F2 serve two txn types; no runtime gate; assignment can happen at context assembly	The cleaner target state. Requires the vault response before classification, which the target architecture already does

Why this document builds on reading A

Reading C is where this should end up, and it simplifies assignment considerably. But it requires credential resolution to move ahead of classification, which is a change to the ingress path and not in scope for this experiment. Reading A works today, is what the sketch describes, and forces a question worth answering anyway: where in a flow can an experiment legitimately be evaluated. §C1 answers it, and the answer holds under reading C as well.

The two new strategies

Two strategies are being deployed, one at each experimentable child of D4. Each is a registered artefact with a population, a fallback and an expiry — not a code change.

```
strategy Cred_PreferMerchantFPAN_v1
  decision_point D4.1
  class DETERMINISTIC
  population txn_type = US_ECOM_CREDIT_CIT
    and credential_set.count > 1
    and credential_set ANY (type = FPAN,
                           scope = MERCHANT)
  definition prefer merchant-scoped FPAN over network token
    when both are present and active
  fallback Cred_PreferToken_v1 ← the incumbent
  requires credential_set with scope and lifecycle state
  guardrails inherited from the experiment
  owner credential.product
  review_date +90d expiry +180d

strategy Attempt_Strategy2_v1
  decision_point D4.2
  class PARAMETRIC
  population txn_type = US_ECOM_CREDIT_CIT
    and credential_set.count > 1
  definition retry strategy 2 – credential substitution
    before processor substitution, and a longer
    scheduled window on issuer-state declines
  fallback Attempt_Standard_v1 ← the incumbent
  requires canonical decline class, credential lifecycle
  owner routing.product
  review_date +90d expiry +180d
```

One assumption to confirm

The sketch annotates the FPAN strategy with the word *merchant*. This document reads that as the credential **scope** — prefer the merchant-scoped FPAN over the network token, rather than a PSP-scoped or customer-scoped credential. If it instead meant "merchant-configured" or "merchant-elected", the population predicate and the strategy class both change, and it becomes a **Cred_Directed** variant rather than a deterministic preference. Worth thirty seconds in the review.

F3 as sketched — finding 1

The new flow is drawn as F3 carrying both new strategies:

```
F3  D1 · D2 · D3 · D4 · D4.1 · D4.2 · D5 · D6
    D4.1 → Cred_PreferMerchantFPAN_v1      ← differs from F2
    D4.2 → Attempt_Strategy2_v1           ← differs from F2

    baseline_ref  F2_Multi_v6
    diff_count    2                        ← does not compile
```

Finding 1 – the experiment cannot answer its own question

F3 differs from F2 at two decision points, so the experiment measures the pair. If it wins by 37 bps, nothing in the data says whether the credential change produced 37, the retry change produced 37, or the credential change produced 55 and the retry change cost 18. If it loses, the same ambiguity applies in reverse and a good change gets discarded along with a bad one. The experiment is named **Token Performance** — a question exclusively about D4.1 — and as drawn it cannot answer that question at any confidence.

This is what the `diff_count = 1` rule exists to catch, and it is caught at registration rather than at readout, which is the only useful time to catch it.

Four ways to fix it

Option	Design	Answers	Cost
1 — Split by layer	F3 differs at D4.1 only. F4 differs at D4.2 only. Two experiments, two layers, independent salts, run concurrently	Both main effects, cleanly. Interaction tested at readout	Two registrations. Interaction analysis before either promotes
2 — Three arms	Control F2; arm A = credential only; arm B = credential + retry	Credential effect, and the combined effect. Not the retry effect alone	One experiment, three arms, more traffic than two
3 — 2×2 factorial	Control; credential only; retry only; both	Both main effects and the interaction, in one design	Four arms. Roughly 2× the traffic of option 1 for the same power per effect
4 — Sequence	Credential experiment first. Promote or reject. Then retry against the new baseline	Both, one at a time, each against a current baseline	Calendar time — two full cycles instead of one

Recommended — option 1

The stated hypothesis is about credential form. The retry change is a separate question the experiment name does not even mention, and bundling it is how it got here. Two experiments on two layers cost one extra registration and run at the same time, so there is no calendar penalty against option 4. Option 3 is the right design only if the real question is whether the *combination* beats either change alone — and nobody has asked that yet.

One caveat that must be carried into the readout: the credential arm changes the decline mix that the retry arm then operates on. Orthogonal assignment does not make the effects orthogonal. The interaction term is tested at readout, and a significant interaction blocks both promotions until they are analysed together.

F3 and F4 compiled

Flow	Role	Baseline	D4.1	D4.2	diff	Differs at
F2_Multi_v6	BASELINE	—	Cred_PreferToken_v1	Attempt_Standard_v1	—	—
F3_MultiFPAN_v1	VARIANT	F2_Multi_v6	Cred_PreferMerchantFPAN_v1	Attempt_Standard_v1	1	D4.1
F4_MultiRetry2_v1	VARIANT	F2_Multi_v6	Cred_PreferToken_v1	Attempt_Strategy2_v1	1	D4.2

Every other binding — D1 through D4, D5, D6 — is identical across all three. That is not a convention; it is what the compiler verifies, and it is what makes the pre-assignment path safe to share.

What the compiler checks at registration

- ✓ diff_count = 1 against the declared baseline
- ✓ the differing node is marked experimentable
- ✓ every decision point in the graph has a binding — no gaps
- ✓ each bound strategy's population contains the flow's txn type
- ✓ each strategy names a fallback, and the chain terminates
- ✓ naming conforms — no sequential names
- ✓ expiry and review_date present
- ✓ the pre-assignment path is bit-identical across arms ← §C1
- ✓ the bundle signs

The assignment point

Assignment happens at the divergence point — the last node common to every arm, immediately before the first node where they differ. Not at context assembly.

The sketch places the experiment evaluation after D4. That is correct, and it corrects the Experimentation PRD, which says assignment happens at context assembly. Context assembly is the right place only when every eligible transaction will reach the treatment. Here, most will not.

Why not at context assembly

IF ASSIGNED AT CONTEXT ASSEMBLY (before D1)

10,000 transactions assigned



dilution: 62% of the test arm carries no treatment.
The measured effect is diluted by the same proportion, so the experiment needs roughly $1/0.38^2 \approx 6.9\times$ the traffic to detect the same true effect. It also reports a smaller number than the true effect, which is how a good change gets rejected.

IF ASSIGNED AT THE DIVERGENCE POINT (after D4, before D4.1)

10,000 transactions



The general rule

assignment_point must satisfy all three:

- 1 AT OR AFTER the last fact any eligibility check depends on here: the vault response (D2) and the gate outcome (D4)
- 2 AT OR BEFORE the divergence point — the first node where the arms differ here: D4.1
- 3 Every node BEFORE the assignment point is bit-identical across all arms

Condition 3 is the one that is easy to lose and fatal to lose. If any node before assignment differed between arms, the population being assigned would already have been shaped by the treatment, and the comparison would be between two different populations rather than two treatments. The compiler verifies it by diffing the graphs up to the assignment node.

For this experiment:

last eligibility fact	D4 outcome	
divergence point	D4.1	
assignment point	between D4 and D4.1	✓ valid
pre-assignment path	D1 · D2 · D3 · D4 — identical in F2, F3 and F4	✓ verified

A correction to the PRD

The Experimentation PRD states that assignment happens once at context assembly and is carried forward. That is right about *once* and *carried forward*, and wrong about *where*. The correct statement is that assignment happens once, at the divergence point, and is carried forward from there — including across retries of the same transaction. Where the divergence point happens to be the first node, that is context assembly; here it is not. The PRD should be amended.

Transaction selection — finding 2

The sketch defines selection as: *vault response has more than one credential, and one of them is a COF token*. As a predicate:

```
AS SKETCHED
credential_set.count > 1
AND credential_set ANY (type = NETWORK_TOKEN AND cof = true)
```

Finding 2 – part of the test arm cannot receive the treatment

The variant's treatment is *prefer the merchant-scoped FPAN*. The predicate does not require an FPAN to be present. A customer with two network tokens and no FPAN satisfies it, gets assigned, and then D4.1 falls back to the incumbent because the treatment credential does not exist. Those transactions sit in the test arm having received the control treatment, which dilutes the effect and drags treatment fidelity down — and fidelity below the floor blocks promotion regardless of what the primary metric says.

Eligibility must require everything the treatment needs, not just everything the hypothesis mentions.

```
CORRECTED
credential_set.count > 1
AND credential_set ANY (type = NETWORK_TOKEN
                        AND cof = true
                        AND lifecycle = ACTIVE)
AND credential_set ANY (type = FPAN
                        AND scope = MERCHANT
                        AND lifecycle = ACTIVE)    ← added
AND customer_ref IS NOT NULL                    ← added, §C3
AND txn_type = US_ECOM_CREDIT_CIT

Lifecycle clauses added on both: a suspended or expired
credential is present in the set and not usable, and
"present" was doing the work of "usable" in the original.
```

The full eligibility ladder

Selection is one of eight checks. Any false sends the transaction to F2 with the reason recorded — never silently.

Check	Fails when	Reason code
Population predicate	The corrected predicate above is not satisfied	POPULATION_MISS
Treatment availability	No active merchant-scoped FPAN in the set	MISSING_TREATMENT_CREDENTIAL
Merchant opt-out	Merchant has declined participation	MERCHANT_OPTED_OUT
Strategic account	Merchant on the named exclusion list	STRATEGIC_ACCOUNT
Permanent holdout	Customer falls in the reserved 2%	HOLDOUT
Migration soak	Transaction is in a cutover slice	MIGRATION_SOAK
Layer occupancy	Another active experiment holds this (layer, segment)	LAYER_OCCUPIED
Autonomy floor	Merchant autonomy below the level the variant requires	AUTONOMY_BELOW_REQUIRED

Unit of randomisation — finding 3

The sketch does not state one. The obvious choice is the transaction, and it is wrong here.

Candidate unit	Assessment
Transaction	Wrong. The same customer would flip between FPAN and network token across consecutive transactions. Issuers observe credential-form changes on a stored credential and treat them as a risk signal, so the arms contaminate each other through the issuer's own model. It also destroys the token lifecycle signal the experiment is partly about
Credential	Wrong. The decision is about a <i>set</i> — which of several credentials to present. Randomising on one member of the set does not define an arm for the decision
Customer	Correct. The credential set belongs to the customer; the decision is stable for that customer; issuer exposure is consistent; token lifecycle stays coherent
Merchant	Too coarse. Merchant-level effects would swamp the credential effect and the sample would collapse to the number of merchants

```
unit_of_randomisation  customer_ref
bucket = hash(customer_ref || salt_cred) mod 10000
```

Consequence, and it is why the predicate gained a clause: guest checkouts have no customer_ref and cannot be assigned stably. They are excluded — POPULATION_MISS — rather than assigned on a fallback key. A mixed-unit design is worse than a smaller one.

What 95/5 costs — finding 4

The sketch sets control F2 at 95% and test F3 at 5%. That is a defensible risk posture and an expensive power posture, and the two pull in opposite directions. Sample sizes below are computed at 80% power, 5% two-sided, against an illustrative 87.0% baseline transaction approval rate — substitute the real figure for the eligible population before committing.

Detect	Split	Test arm	Control arm	Total eligible	vs 50/50
25 bps	95 / 5	149,510	2,840,697	2,990,207	5.3×
	90 / 10	157,816	1,420,348	1,578,165	2.8×
	80 / 20	177,544	710,174	887,718	1.6×
	50 / 50	284,070	284,070	568,139	1.0×
40 bps	95 / 5	58,402	1,109,647	1,168,050	5.3×
	90 / 10	61,647	554,824	616,471	2.8×
60 bps	95 / 5	25,957	493,176	519,133	5.3×
	90 / 10	27,399	246,588	273,987	2.8×

Calendar time at 25 bps, by eligible daily volume

Eligible / day	95 / 5	90 / 10	80 / 20	50 / 50
20,000	150 d	79 d	44 d	28 d
50,000	60 d	32 d	18 d	11 d
100,000	30 d	16 d	9 d	6 d
250,000	12 d	6 d	4 d	2 d

Finding 4 — and the shape of the trade

Note the direction of the numbers. Moving from 95/5 to 50/50 *increases* what the test arm needs, from 150k to 284k, while *halving* total traffic from 3.0M to 568k. An imbalanced split minimises exposure to the variant, which is exactly what you want while the guardrails are unproven — and it maximises the total traffic consumed, which is exactly what you do not want once they are.

The eligible population here is narrow: multi-credential, COF token present, FPAN present, US ecom credit CIT, customer identified. If that is 20,000 a day, 95/5 is a five-month experiment for a 25 bps effect. Somebody should put a number on the eligible population before the split is fixed.

The recommended shape — a two-phase ramp

- PHASE 1 SOAK

95 / 5, 7 days or 40,000 eligible, whichever is later
purpose: guardrails, fidelity, no harm
NOT part of the primary analysis
- PHASE 2 POWER

90 / 10, run to the pre-registered n
purpose: the effect estimate
- Both phases are pre-registered before launch, with the phase boundary declared. The primary analysis uses phase 2 only.
- Why the phases are separated rather than blended: a mid-flight allocation change makes a fixed-sample analysis invalid unless it is modelled. Pooling a 95/5 period with a 90/10 period without time stratification produces a sample-ratio mismatch that most analysis tooling will flag and some will silently absorb. Declaring two phases and analysing the second is simpler than doing the stratification correctly, and it loses very little — phase 1 is small by construction.

The experiment record

The sketch's five fields, completed to what the registry requires. Fields carried straight from the sketch are marked.

experiment	EXP_TokenPerformance_FPANvsToken			
experiment_id	exp_01JQ8K...			
name	Token Performance		← sketch	
owner	credential.product			
hypothesis	"For US ecom credit CIT transactions where the vault returns both an active COF network token and an active merchant-scoped FPAN, preferring the FPAN improves transaction approval by at least 25 bps without raising fraud or dispute proxies."			
txn_type	US_ECOM_CREDIT_CIT			
decision_point	D4.1			
layer	credential_selection			
layer_salt	salt_cred			
segment	us_ecom_credit_multicred		← controlled vocabulary	
assignment_point	post-D4, pre-D4.1		← §C1	
unit_of_randomisation	customer_ref		← §C3	
arms				
control	F2_Multi_v6	phase 1	95%	← sketch
		phase 2	90%	
variant	F3_MultiFPAN_v1	phase 1	5%	← sketch
		phase 2	10%	
holdout	reserved 2% off the top, never assigned			
population	see §C2 corrected predicate			
exclusions	merchant_opt_out · strategic_account · holdout · migration_soak · autonomy_floor			
primary_metric	transaction_approval_rate			
	denominator: transactions with ≥1 attempt			
mde	25 bps			
power	0.80 at alpha 0.05 two-sided			
sample_size	157,816 variant / 1,420,348 control (phase 2)			
stopping_rule	FIXED_SAMPLE – no peeking at the primary			
guardrails	fraud_proxy	+15% rel	abort	
	dispute_proxy	+10% rel	halt	
	cost_of_acceptance	+10 bps	abort	
	per_merchant_harm	beyond CI	abort	
	token_lifecycle_events	+20% rel	halt	
	treatment_fidelity	<95%	block promote	
	constraint_violation	any	abort	
outcome_horizons	approval	seconds		
	cost	2 days		
	fraud	30 days		
	dispute	120 days		
attribution_baseline	F2_Multi_v6			
interaction_layers	attempt_strategy (EXP_Retry2, if concurrent)			
timeframe	phase 1 start	2026-10-06	← sketch	
	phase 2 start	on soak clearance		
	planned_end	on sample_size reached		
	hard_expiry	2027-01-31		
decision	pending			

One guardrail that is specific to this experiment

token_lifecycle_events is not in the standard set. Moving traffic off network tokens and onto FPAN reduces token usage, and sustained non-use affects token lifecycle state and issuer-side token health. An experiment that improves approval while quietly degrading the token estate has borrowed from a capability the platform will want later. Whether +20% is the right threshold is a question for the credential team; that it needs a threshold is not.

What registration rejects

Run against the registry as sketched, before any of the corrections above.

Gate	Sketch	Result
Flow single difference	F3 differs at D4.1 and D4.2	REJECT — <code>diff_count = 2</code>
Treatment availability in population	FPAN not required by the predicate	REJECT — treatment not guaranteed deliverable
Unit of randomisation	Not declared	REJECT — required field
Primary metric	Not declared. "Token Performance" is a name, not a metric	REJECT — required, exactly one
Sample size and power	Not declared	REJECT — required with the MDE it detects
Stopping rule	Not declared	REJECT — FIXED_SAMPLE or SEQUENTIAL
Guardrails	None	REJECT — minimum set required
Hard expiry	Timeframe left blank	REJECT — required
Layer and segment	Not declared	REJECT — uniqueness cannot be checked without them
Strategy fallbacks	Not declared on either new strategy	REJECT — never implicit
Attribution baseline	Implicit — F2	WARN — must be declared, not inferred
Merchant scope	"* any or specific" — undecided	WARN — resolves to all, minus exclusions
Assignment point	Post-D4	PASS — and verified against the divergence point
Control weight	95% with 5% variant	PASS — but see §C4 on duration
Naming	Descriptive, not sequential	PASS

Ten rejections, two warnings, three passes. None of the ten is a judgement call — each is a missing required field or a mechanical rule, which is the point of making them mechanical. The registration form does this work so the review meeting does not have to.

T1 — single credential. Never reaches the experiment.

T2 — eligible, assigned to control.

T3 — eligible, assigned to test. The treatment lands.

customer	cus_C08	·	amount 12,500
D4 gate	count = 3 → YES branch		
— ASSIGNMENT POINT			
eligibility	all 8 checks pass		
bucket	hash(cus_C08 salt_cred) = 9731		
	9500-9999 → variant (phase 1, 5%)		
arm	variant · propensity 0.05		
flow	F3_MultiFPAN_v1		
D4.1 credential	Cred_PreferMerchantFPAN_v1 → FPAN		
	considered: [NETWORK_TOKEN, FPAN,		
	ACQUIRER_TOKEN]		
	rejected: NETWORK_TOKEN		
	NOT_SELECTED_BY_STRATEGY		
	ACQUIRER_TOKEN		
	NOT_SELECTED_BY_STRATEGY		
	fallback_would_have_been: NETWORK_TOKEN		
	← the counterfactual		

		attribution needs
D4.2 attempt	Attempt_Standard_v1	← unchanged from F2
D5 · D6	as F2	← unchanged from F2
outcome	APPROVED	
fidelity	treatment delivered	

T4 — multi-credential, but the treatment does not exist.

```
customer cus_D55 · amount 2,100

D4 gate          count = 2 → YES branch
                  { network_token · cof · ACTIVE }
                  { network_token · wallet · ACTIVE }
                  no FPAN in the set

— ASSIGNMENT POINT —
eligibility      treatment availability FAIL
reason          MISSING_TREATMENT_CREDENTIAL
arm             none — not assigned
flow            F2_Multi_v6 (baseline)

D4.1            Cred_PreferToken_v1 → NETWORK_TOKEN
outcome         APPROVED
```

This is finding 2 working as intended. Under the sketch's original predicate this transaction would have been assigned, landed in the variant 5% of the time, fallen back at D4.1, and counted as a treated observation that received control. Under the corrected predicate it is excluded before assignment, with the reason on the record.

T5 — eligible population, permanent holdout.

```
customer cus_E22 · amount 6,400

D4 gate          count = 2 → YES branch

— ASSIGNMENT POINT —
eligibility      permanent_holdout FAIL
reason          HOLDOUT
arm             none — not assigned, and never will be
flow            F2_Multi_v6 (baseline)

outcome         DECLINED · DO_NOT_HONOR
```

The holdout is not an exclusion in the ordinary sense. These transactions are the measuring stick for the cumulative effect of every strategy the platform has promoted, which is why they are excluded from exploration as well, and why the 2% cannot be created retroactively.

T6 — assigned to test, treatment not delivered.

```
customer cus_F73 · amount 15,750

D4 gate          count = 2 → YES branch
                  { network_token · cof · ACTIVE }
                  { fpan · merchant · ACTIVE } at D2

— ASSIGNMENT POINT —
eligibility      all 8 checks pass
bucket          9884 → variant
arm             variant · propensity 0.05
flow            F3_MultiFPAN_v1

D4.1 credential  Cred_PreferMerchantFPAN_v1
                  FPAN lifecycle re-read: SUSPENDED
                  (issuer-driven, between D2 and D4.1)
                  strategy cannot produce its action
                  → FALLBACK Cred_PreferToken_v1
                  → NETWORK_TOKEN
                  regime: FALLBACK
                  fidelity: treatment_delivered = false
outcome         APPROVED
```

The transaction STAYS IN THE VARIANT ARM. It is not dropped and not reassigned. Dropping it would bias the result toward the moments when the credential estate happened to be healthy — which is not the population the strategy would run in. It is flagged, and the arm's fidelity is reported beside the effect.

The records

T3 in full — the variant transaction where the treatment landed. Everything else in this document is a view over records shaped like this one.

```
{
  "attempt_id": "att_01JQ8M4KX2",
  "transaction_id": "txn_01JQ8M4KWZ",
  "txn_type": "US_ECOM_CREDIT_CIT",
  "merchant_id": "mer_4471",
  "customer_ref": "cus_C08",

  "exec_id": "F3_MultiFPAN_v1",
  "flow_version": "1.0",
  "flow_baseline_ref": "F2_Multi_v6",

  "experiment": {
    "experiment_id": "EXP-TokenPerformance_FPANvsToken",
    "assignment_id": "asg_01JQ8M4KX1",
    "layer": "credential_selection",
    "segment": "us_ecom_credit_multicred",
    "assignment_point": "post_D4",
    "unit_type": "customer_ref",
    "unit_id": "cus_C08",
    "bucket": 9731,
    "arm": "variant",
    "propensity": 0.05,
    "phase": 1
  },

  "eligibility": { "eligible": true, "excluded_by": null,
                  "checks_passed": 8 },

  "credential_set_at_D2": [
    { "id": "cred_9f2", "type": "NETWORK_TOKEN", "cof": true,
      "scope": "PSP", "lifecycle": "ACTIVE" },
    { "id": "cred_3b7", "type": "FPAN", "cof": true,
      "scope": "MERCHANT", "lifecycle": "ACTIVE" },
    { "id": "cred_c41", "type": "ACQUIRER_TOKEN", "cof": false,
      "scope": "PSP", "lifecycle": "ACTIVE" }
  ],

  "decisions": [
    { "decision_point": "D4",
      "outcome": "MULTI", "regime": "POLICY", "propensity": 1.0 },

    { "decision_point": "D4.1",
      "strategy": "Cred_PreferMerchantFPAN_v1", "version": "1.0",
      "action": "cred_3b7",
      "regime": "POLICY", "propensity": 1.0,
      "considered": ["cred_9f2", "cred_3b7", "cred_c41"],
      "rejected": [
        { "action": "cred_9f2",
          "reason_code": "NOT_SELECTED_BY_STRATEGY" },
        { "action": "cred_c41",
          "reason_code": "NOT_SELECTED_BY_STRATEGY" }
      ],
      "fallback_would_have_been": "cred_9f2",
      "duration_ms": 0.7 },

    { "decision_point": "D4.2",
      "strategy": "Attempt_Standard_v1", "version": "1.0",
      "action": "NO_ACTION_REQUIRED", "regime": "HEURISTIC",
      "propensity": 1.0,
      "fallback_would_have_been": "NO_ACTION_REQUIRED" },

    { "decision_point": "D5",
      "strategy": "Auth_RiskAdaptive_v3", "action": "NONE",
      "regime": "LEARNED", "propensity": 0.91,
      "fallback_would_have_been": "NONE" },

    { "decision_point": "D6",
      "strategy": "Route_Balanced_v7", "action": "proc_a",
      "regime": "LEARNED", "propensity": 0.64,
      "considered": ["proc_a", "proc_b", "proc_d"],
      "rejected": [
        { "action": "proc_b", "reason_code": "HEALTH_DEGRADED" },
        { "action": "proc_d", "reason_code": "HIGHER_EXPECTED_COST" }
      ],
      "fallback_would_have_been": "proc_b" }
  ],

  "artefact_versions": {
```

```

    "constraint_set": "v311", "bin_snapshot": "v52",
    "routing_table": "v184", "cost_table": "v77",
    "models": { "fraud": "v19", "route_auth": "v22" }
  },

  "fidelity": { "treatment_delivered": true, "fallbacks": 0 },

  "result": { "outcome": "APPROVED", "auth_code": "A17R4C" },
  "outcome_joins": {
    "approval": { "status": "final", "value": true },
    "cost":      { "status": "pending", "closes_at": "2026-10-09" },
    "fraud":     { "status": "pending", "closes_at": "2026-11-06" },
    "dispute":   { "status": "pending", "closes_at": "2027-02-04" }
  }
}

```

T1, for contrast — the absent block

```

{
  "attempt_id": "att_01JQ8M4KA7",
  "txn_type": "US_ECOM_CREDIT_CIT",
  "exec_id": "F1_Single_v4",
  "decisions": [
    { "decision_point": "D4", "outcome": "SINGLE",
      "regime": "POLICY", "propensity": 1.0 },
    { "decision_point": "D5", ... },
    { "decision_point": "D6", ... }
  ],
  "result": { "outcome": "APPROVED" }
}

```

No "experiment" key. No "eligibility" key. The transaction is not in the experiment's denominator at any level, because it never reached the point where membership is decided.

How the data appears

Txn	Gate	ExecID	Eligible	Arm	Prop.	Credential used	Fidelity	In readout
T1	SINGLE	F1_Single_v4	—	—	—	network token	—	No
T2	MULTI	F2_Multi_v6	true	control	0.95	network token	✓	Yes
T3	MULTI	F3_MultiFPAN_v1	true	variant	0.05	FPAN	✓	Yes
T4	MULTI	F2_Multi_v6	false	—	—	network token	—	No
T5	MULTI	F2_Multi_v6	false	—	—	network token	—	Holdout only
T6	MULTI	F3_MultiFPAN_v1	true	variant	0.05	network token	✗	Yes

Three things this table is designed to make visible

- **T4 and T5 both show eligible: false and both run F2 — but they are not the same.** T4's exclusion is a property of the transaction and could change tomorrow. T5's is permanent and by design. Only the reason code separates them, which is why the reason is recorded rather than just the verdict.
- **T6 is in the readout with the control treatment.** Its ExecID says F3, its credential says network token. Any analysis that groups by credential used rather than by arm assigned will silently move T6 into the control group and bias the estimate toward zero. Analysis groups by arm, always — the intention-to-treat population — and fidelity is reported separately.
- **T1 has no experiment block at all.** A transaction that never reached the assignment point is not a zero in the experiment; it is outside it. Storing an eligibility verdict for it would inflate every denominator in the readout by the size of the single-credential population, which here is about 62%.

Readout at day 37

EXP_TokenPerformance_FPANvsToken		status		READOUT
population	US_ECOM_CREDIT_CIT	multi-credential · active COF token AND active merchant FPAN present		
unit	customer_ref	layer	credential_selection	
baseline	F2_Multi_v6	assignment	post-D4	
PHASE 1	soak	days 1-7	95/5	364,200 eligible
		excluded from the primary analysis		
PHASE 2	power	days 8-37	90/10	1,578,000 eligible
		control 1,420,200 · variant 157,800		
TREATMENT FIDELITY				
	control	99.94%		
	variant	97.22%		
	floor	95.00%		
		← 2.78% fell back at D4.1		
		PASS, but see note		
PRIMARY transaction approval rate				
	control	87.31%		
	variant	87.68%		
	delta	+37 bps	CI [+20, +54]	PASS
ITT estimate. Adjusted for fidelity the implied treated effect is +38 bps; the readout reports the ITT figure.				
GUARDRAILS				
	fraud proxy (30d)	+0.4% rel	CI [-1.9, +2.7]	within
	dispute proxy	+2.1% rel		within
		confirmed horizon 2027-02-04		pending
	cost of acceptance	+2.1 bps		within
		FPAN carries no token benefit		
	per-merchant harm	none beyond CI at ≥5,000		PASS
	token lifecycle events	+11% rel	(threshold +20%)	within
		trending up – see note		
	latency	no change		PASS
INTERACTION				
	concurrent layer	attempt_strategy – EXP_Retry2 live from d12		
	interaction term	+3 bps	CI [-14, +20]	not significant
both promotions may proceed independently				
CALIBRATION				
	challenger predicted +34 bps where champion and challenger agreed; observed +36 bps. calibration error 2 bps.			
DECISION PROMOTE to 50% exposure. Full ramp and baseline recompile held until the 120-day dispute horizon closes on 2027-02-04.				

Two things in this readout that deserve argument, not acceptance

Cost is up 2.1 bps and the guardrail passed anyway. Network tokens carry interchange and fraud-liability benefits that FPAN does not, so moving traffic to FPAN should be expected to cost something. +37 bps of approval against +2.1 bps of cost is a good trade at almost any exchange rate, but it is a trade, and the readout should present it as one rather than as a clean win. If the merchant's posture were MINIMISE_COST this result would read differently.

Token lifecycle events are up 11% and the threshold is 20%. Thirty-seven days is not long enough to know whether that plateaus or keeps climbing. Sustained non-use degrades token health and the second-order effects land months later, well outside this readout's window. Promotion to 50% doubles the pressure. The credential team should own a view on this before the full ramp, not after.

What the holdout says separately

The 2% holdout is not an arm of this experiment and does not appear above. It answers a different question, on its own schedule:	
holdout approval	86.94%
full population	87.55% (control and variant, weighted)
cumulative delta	+61 bps
That +61 bps is the aggregate effect of every strategy promoted across all decision points since the holdout was reserved – not this experiment's contribution. It is the only number that answers "what has the decision platform been worth", and it exists only because the 2% was set aside before any of it started.	

Review agenda

Twelve items. The first four block registration, the next four block launch, and the last four block the full ramp. Each has a named owner type rather than a name, because the point of the meeting is to attach names.

Blocks registration

#	Decision	Owner
1	Confirm D4.1 is credential selection and D4.2 is attempt strategy, and freeze the identifiers	Architecture
2	Accept option 1 — split F3 into two single-difference flows on two layers — or choose another of the four	Product
3	Confirm <i>merchant</i> in the FPAN strategy means credential scope, not merchant configuration	Credential product
4	Adopt the corrected eligibility predicate, including the FPAN-present and lifecycle clauses	Product + credential

Blocks launch

#	Decision	Owner
5	Put a number on the eligible population per day. Everything in §C4 is hypothetical until this exists	Data science
6	Choose the split, given that number and the duration table. Two-phase ramp or a single fixed split	Product + data science
7	Confirm <i>customer_ref</i> as the unit, and that guest checkouts are excluded rather than assigned on a fallback key	Data science
8	Set the <i>token_lifecycle_events</i> threshold, or decide it is not a guardrail	Credential product

Blocks the full ramp

#	Decision	Owner
9	Amend the Experimentation PRD: assignment happens at the divergence point, not at context assembly	Architecture
10	Decide whether the 2.1 bps cost increase is acceptable at portfolio level and at merchant level, and whether posture changes the answer	Product + finance
11	Agree the interaction policy when the credential and retry experiments overlap — who reads the interaction term, and what blocks on it	Data science
12	Reserve the permanent holdout if it does not exist yet. It cannot be created retroactively, and every number in §D4's final block depends on it	Data science

The one to settle first

Item 5. The split, the duration, the phase design and whether this experiment is worth running at all are all downstream of how many transactions a day have two credentials including both an active COF token and an active merchant-scoped FPAN. If that number is 20,000, a 25 bps effect takes five months at 95/5 and the conversation becomes whether to widen the population or accept a larger MDE. If it is 200,000, none of this matters and the experiment can run at 95/5 for a fortnight. It is a query, not a decision, and it should be run before the meeting rather than in it.

Not on this list

Propensity and fallback logging on the current engine. That is not a decision for this review — it is the prerequisite for every learned strategy that comes after this one, it changes no behaviour, and it cannot be backfilled. It should already be in flight.